

Grade 7
Unit 12
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

1. To find the volume of a cylinder, multiply the _____ of the base by the _____ of the cylinder.

☐ diameter
☐ radius
☒ height

☐ area
☐ circumference

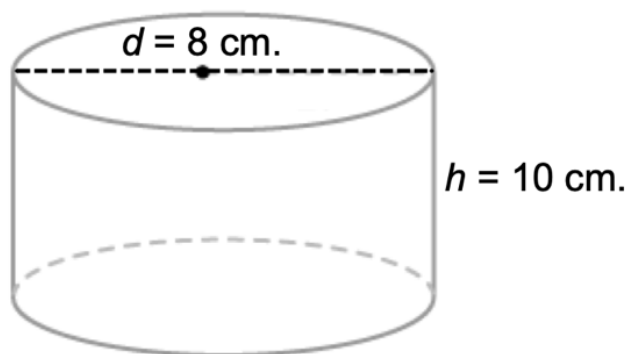
2. Which formula(s) represent the volume of a right cylinder?

- ☐ $V = r^2 \cdot h$, where r is the radius and h is the height.
☒ $V = \pi r^2 \cdot h$, where r is the radius and h is the height.
☒ $V = B \cdot h$, where B is the area of the base and h is the height.
☐ $V = C \cdot h$, where C is the circumference and h is the height.

A diagram of a right circular cylinder is shown with the diameter of 8 centimeters (cm.) and height of 10 cm for questions 3-4.

3. Calculate the area of the base. Leave in terms of π and use the appropriate units.

$$\begin{aligned} r &= 4 & A &= \pi r^2 \\ & & A &= \pi(4)^2 \\ & & A &= 16\pi \text{ cm}^2 \end{aligned}$$

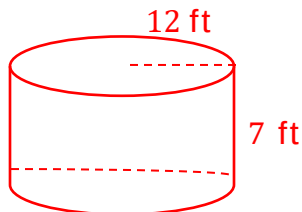


4. Calculate the volume of the cylinder. Leave in terms of π and use the appropriate units.

$$\begin{aligned} V &= \pi r^2 h & V &= 16\pi \cdot 10 \\ V &= 160\pi \text{ cm}^3 \end{aligned}$$

For questions 5-6, consider a cylinder with a radius of 12 feet (ft.) and height of 7 ft.

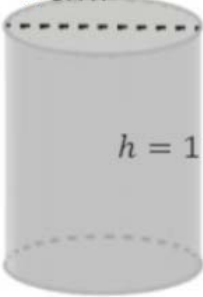
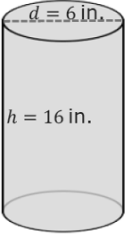
5. Sketch the cylinder and label its dimensions.



6. Calculate the volume of the cylinder using $\frac{22}{7}$ for π . Include appropriate units.

$$\begin{aligned} V &= \frac{22}{7} \cdot 12^2 \cdot 7 & V &= 3,168 \text{ ft}^3 \end{aligned}$$

Four right circular cylinders are shown and specific measurements are given. The measurements include either the radius, r , diameter, d , or height, h . For each cylinder, find the volume. Use 3.14 to approximate π , round to the nearest tenth, and include appropriate units.

Cylinder	Volume
7. $h = 5$ units $r = 2$ units 	$V = \pi r^2 h$ $V = 3.14 \cdot 2^2 \cdot 5$ $V = 62.8 \text{ units}^3$
8. $d = 8$ cm.  $h = 11$ cm.	$V = \pi r^2 h$ $V = 3.14 \cdot 4^2 \cdot 11$ $V = 552.6 \text{ cm}^3$
9. $h = 1.7$ m.  $r = 3$ m.	$V = \pi r^2 h$ $V = 3.14 \cdot 3^2 \cdot 1.7$ $V = 48.0 \text{ m}^3$
10.  $d = 6$ in. $h = 16$ in.	$V = \pi r^2 h$ $V = 3.14 \cdot 3^2 \cdot 16$ $V = 452.2 \text{ in}^3$